

Kinematic and Constraint Basis of the Linkage Simulator

Implementation-derived technical note

1 Scope and conventions

The application is a deterministic planar kinematics simulator with a bounded quasi-static gravity estimate, not a dynamic, clinical, or musculoskeletal model. The serial finger abstraction follows established hand kinematic modelling practice [1, 2]. World coordinates use millimetre-like units, angles use radians internally, and positive angles are counter-clockwise. A rigid link has pose $(\mathbf{p}, \theta)$, and an arbitrary local point $\mathbf{q}$ maps to world space as

$$\mathbf{x}(\mathbf{p}, \theta, \mathbf{q}) = \mathbf{p} + R(\theta)\mathbf{q}, \quad R(\theta) = \begin{bmatrix} \cos \theta & -\sin \theta \\ \sin \theta & \cos \theta \end{bmatrix}. \quad (1)$$

The rectangular length and width are rendering and editing dimensions; joint attachments are not required to lie at $\pm L/2$. Geometry and mechanism solving have no DOM or Canvas dependency, and rendering does not mutate state.

2 Independent digit workspaces

The application instantiates independent simulation states for D2 through D5 and renders only the active tab. Servo angle, animation direction, linkage edits, contactors, joint-limit status, and static-moment results therefore persist independently for each digit. A shared hand-size factor h applies to all four anatomical models. Neutral initialization uses rounded between-digit scales

$$(s_2, s_3, s_4, s_5) = (1.00, 1.11, 1.03, 0.83), \quad (2)$$

consistent with reported young-adult mean digit lengths of approximately 63.9, 70.7, 65.5, and 53.3 mm, respectively [3]. For base D2 segment length $L_{i,2}$, digit d is initialized with $L_{i,d} = h s_d L_{i,2}$. The linkage template is scaled by s_d at creation; subsequent hardware edits remain local to one digit.

3 Dorsal mounting and actuator convention

Let $\mathbf{G}$ be a point on the dorsal ground plane, with tangent $\mathbf{t}_g$ and dorsal normal $\mathbf{n}_g$. The editable servo-ground offset h_g places the servo axis at

$$\mathbf{A} = \mathbf{G} + h_g \mathbf{n}_g. \quad (3)$$

The fixed dorsal rail has length L_g and stored orientation α_g , so its second pivot is

$$\mathbf{D} = \mathbf{A} + L_g (\cos \alpha_g, \sin \alpha_g). \quad (4)$$

The rail is identified through the servo’s referenced revolute joint rather than a reserved link name. That revolute must connect the driven link to world or a fixed link, and its fixed-side world attachment must coincide with $\mathbf{A}$; unsupported or mismatched mounts are rejected during graph construction.

The primary servo command ϕ preserves the original application semantics: it is the absolute world orientation of the driven link, not an angle relative to the rail. If $\mathbf{q}_s$ is the driven link’s local point at the servo revolute, the actuator seed is

$$\theta_s = \phi, \quad \mathbf{p}_s = \mathbf{A} - R(\phi)\mathbf{q}_s. \quad (5)$$

The animation advances ϕ between its configured limits and reverses at each bound. Actuator mobility analysis adds the equality $\text{wrap}(\theta_s - \phi) = 0$; passive analysis omits it.

4 Constraint graph and connected components

Every rigid link is a graph body, together with a world body. A fixed link is connected to world and contributes no configuration variables. Each revolute joint is an edge between two link bodies or between a link and world. A zero-range revolute adds a locked relative-angle edge, and the servo contributes an actuator edge referencing its driven revolute joint. A linear-slot edge joins a pin body to a slot body (or world) and contributes the single normal-closure equality derived below. Component discovery is based on adjacency and is independent of IDs and array insertion order.

For each connected component the analysis records its links, joints, actuators, connection to fixed/world geometry, unresolved links, scalar residual count, Jacobian ranks, and local mobilities. A component without a world or fixed connection is reported as free-floating. Its three global rigid motions are retained; it is never silently attached to world. Multiple components are analyzed separately.

Figure 1 shows the update path. Analytic propagation remains the preferred solver. Numerical minimization is reached only for an unresolved subsystem that is still constrained rather than genuinely free.

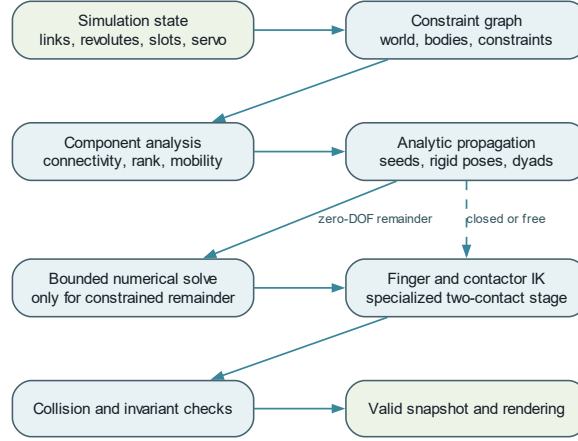


Figure 1: Constraint-graph analysis, mechanism solution, and downstream finger/contact processing.

5 Reference-derived single-digit synthesis architecture

The extended and flexed reference photographs motivate a compact dorsal actuator, a perforated or slotted set of long links, crossed load paths, and a distal output. Perspective, occlusion, and the absence of a dimensional datum do not support a unique reconstruction, so Figure 2 is a proposed synthesis topology rather than a traced measurement. It uses one servo-driven closed chain and one rigid distal coupler point. This is the smallest useful architecture for testing the requested single-contact design; additional loops visible in a physical prototype may later be introduced as load-sharing structure without changing the residual formulation.

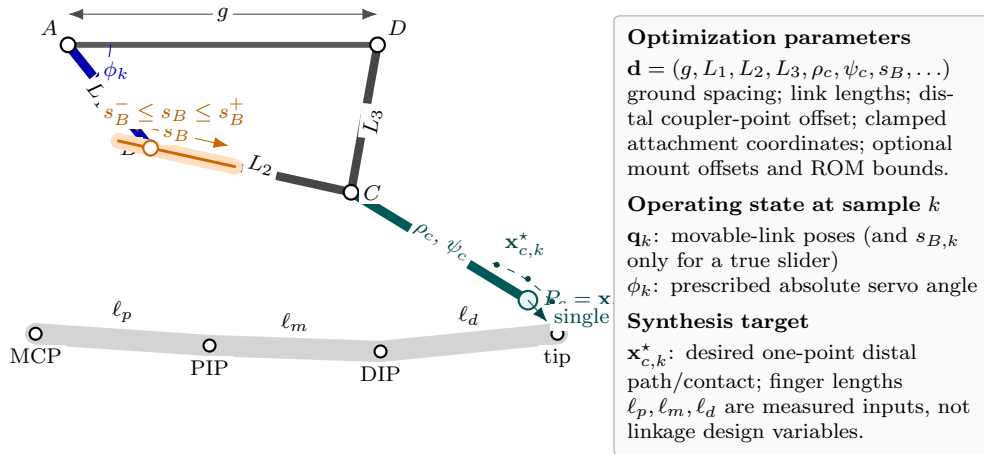


Figure 2: Reference-inspired candidate architecture for one digit and the principal geometry exposed to trajectory optimization. The orange attachment is either clamped at design coordinate s_B or promoted to a bounded runtime slider; those are different mathematical models.

For a clamped perforation or slot, s_B belongs to the design vector $\mathbf{d}$ and remains constant across every sampled pose. The attachment is then an ordinary revolute at a design-dependent local point. For a pin that actually slides during motion, $s_{B,k}$ belongs to the configuration vector $\mathbf{q}_k$ and adds a degree of freedom. Let the slot body have local origin $\mathbf{o}_s$, unit axis $\mathbf{u}_s$, and unit normal $\mathbf{n}_s$. Its world locus and the eliminated-coordinate residual are

$$\mathbf{x}_s(s) = \mathbf{p}_s + R(\theta_s)(\mathbf{o}_s + s\mathbf{u}_s), \quad s^- \leq s \leq s^+, \quad (6)$$

$$r_\perp = (R(\theta_s)\mathbf{n}_s)^\top [\mathbf{x}_p - \mathbf{p}_s - R(\theta_s)\mathbf{o}_s] = 0, \quad (7)$$

$$s = \mathbf{u}_s^\top R(-\theta_s)(\mathbf{x}_p - \mathbf{p}_s) - \mathbf{u}_s^\top \mathbf{o}_s. \quad (8)$$

Equation (7) removes only normal separation; the travel interval is an inequality. Replacing a revolute by this joint therefore removes one scalar equality and generally adds one mobility. A unique commanded pose then needs another independent loop, a contact constraint, or a force/equilibrium law.

Friction is deliberately not inserted into geometric closure. In a later quasi-static stage, a slot reaction with normal magnitude N can satisfy

$$|F_t| \leq \mu_s N \quad (\text{sticking}), \quad F_t = -\mu_k N \operatorname{sign}(\dot{s}) \quad (\text{sliding}). \quad (9)$$

The normal load and the stick-slip mode are not identifiable from positions alone. Reported FFF-polymer friction values vary substantially with material pair, print orientation, surface roughness, load, speed, and lubrication [6, 4, 5]. Consequently the model stores a user-supplied coefficient as metadata but does not label a single PLA or ABS value as a universal default.

6 Implicit design sensitivities and trajectory optimization

Let $\mathbf{d}$ collect the geometry named in Figure 2, and solve one converged mechanism state for each prescribed command ϕ_k :

$$\mathbf{r}(\mathbf{q}_k, \mathbf{d}, \phi_k) = \mathbf{0}. \quad (10)$$

A basic single-contact synthesis objective is

$$\min_{\mathbf{d}} \Phi(\mathbf{d}) = \sum_{k=1}^N w_k \|\mathbf{x}_c(\mathbf{q}_k, \mathbf{d}) - \mathbf{x}_{c,k}^*\|_2^2 + \lambda_L \mathcal{R}_L(\mathbf{d}) + \lambda_M \mathcal{R}_M(\mathbf{d}), \quad (11)$$

subject to closure, ROM, slot-travel, manufacturability, and collision constraints at every sample. $\mathcal{R}_L$ may penalize excessive total link length, while $\mathcal{R}_M$ can preserve clearance from singularity through a conditioning margin. The desired points may be sampled from a measured distal trajectory or generated from the simplified finger model.

On a converged branch with a fixed active set, differentiating Equation (10) gives

$$J_q S + J_d = 0, \quad J_q = \frac{\partial \mathbf{r}}{\partial \mathbf{q}}, \quad J_d = \frac{\partial \mathbf{r}}{\partial \mathbf{d}}, \quad S = \frac{\partial \mathbf{q}}{\partial \mathbf{d}}. \quad (12)$$

The implementation estimates both Jacobians with central differences and solves $J_q S = -J_d$ directly for square full-rank systems or by a full-column-rank least-squares solve for tall systems. The total objective derivative is

$$\frac{d\Phi}{d\mathbf{d}} = \frac{\partial \Phi}{\partial \mathbf{d}} + S^\top \frac{\partial \Phi}{\partial \mathbf{q}}. \quad (13)$$

This is local implicit differentiation, not automatic differentiation through the iterative solver. An underdetermined or rank-deficient J_q , an incompatible overdetermined linearization, a branch switch, a newly active ROM or slot bound, or a stick-slip transition invalidates the ordinary derivative and is reported rather than regularized silently.

7 Residual equations, rank, and local mobility

Each nonfixed planar link contributes $\mathbf{q}_i = (x_i, y_i, \theta_i)$ to the component configuration vector. For a revolute joint j between bodies A and B , with local attachment points $\mathbf{a}_j$ and $\mathbf{b}_j$, the two closure residuals are

$$\mathbf{r}_j(\mathbf{q}) = \mathbf{p}_A + R(\theta_A)\mathbf{a}_j - \mathbf{p}_B - R(\theta_B)\mathbf{b}_j. \quad (14)$$

For a world-side joint the corresponding ground point replaces the first body term. A locked revolute adds the wrapped angular residual

$$r_j^\theta = \ell_\theta \text{wrap}(\theta_B - \theta_A - \bar{\theta}_j), \quad (15)$$

where $\ell_\theta = 50$ mm converts radians to a length-like scale. This characteristic length is comparable to the demonstrator links and avoids combining angular and positional residuals without scaling. The prescribed actuator residual uses the same scale.

For stacked residual vector $\mathbf{r}$, a central finite-difference Jacobian approximates

$$J(\mathbf{q}) = \frac{\partial \mathbf{r}}{\partial \mathbf{q}}. \quad (16)$$

Scaled elimination with named absolute and relative pivot tolerances estimates the rank. The local mobility is

$$M_{\text{local}} = n_q - \text{rank } J. \quad (17)$$

The calculation is performed once without the actuator equality and once with the current command prescribed. Rank, rather than raw equation count, is the informative local diagnostic when constraints are dependent.

At a regular default pose there are six movable links and therefore $n_q = 18$. Seven revolutes contribute 14 positional residuals, and the locked dorsal-anchor joint contributes one angular residual. The passive Jacobian has rank 15, giving $M_{\text{passive}} = 3$. Adding the absolute servo-orientation equality raises the rank to 16 and gives $M_{\text{driven}} = 2$. The classical four-bar subgraph accounts for one passive coordinate and becomes zero-DOF when driven. The two remaining coordinates belong to the middle and distal driver joints and are resolved later by finger/contact constraints.

8 Analytic propagation

The solver first marks fixed links resolved and constructs the servo-driven pose from its referenced joint. It then repeats bounded graph passes until no operation makes progress.

Two-point rigid reconstruction. Suppose local points $\mathbf{q}_1, \mathbf{q}_2$ on one unresolved body have known world positions $\mathbf{x}_1, \mathbf{x}_2$. After checking $\|\mathbf{x}_2 - \mathbf{x}_1\| \simeq \|\mathbf{q}_2 - \mathbf{q}_1\|$, the pose is

$$\theta = \text{atan2}(x_{2y} - x_{1y}, x_{2x} - x_{1x}) - \text{atan2}(q_{2y} - q_{1y}, q_{2x} - q_{1x}), \quad (18)$$

$$\mathbf{p} = \mathbf{x}_1 - R(\theta)\mathbf{q}_1. \quad (19)$$

Coincident local points are degenerate, and incompatible local/world distances are inconsistent. Link length is not changed by reconstruction.

Locked-angle propagation. If one body at a revolute is resolved and $\theta_B - \theta_A = \bar{\theta}_j$ is locked, the other orientation follows directly. Its translation then places its own local attachment at the known world joint. This graph rule replaces the former special dorsal-anchor angle.

Topology-discovered dyads. Two unresolved bodies form an analytic dyad when they share a revolute joint X and each has one other revolute attached to resolved geometry. Let the known outer joint positions be $\mathbf{c}_1, \mathbf{c}_2$. The circle radii come from local attachment geometry, not rectangle length:

$$\rho_1 = \|\mathbf{q}_{X,1} - \mathbf{q}_{O,1}\|, \quad \rho_2 = \|\mathbf{q}_{X,2} - \mathbf{q}_{O,2}\|. \quad (20)$$

The shared point satisfies

$$\|\mathbf{x}_X - \mathbf{c}_1\| = \rho_1, \quad \|\mathbf{x}_X - \mathbf{c}_2\| = \rho_2. \quad (21)$$

The robust circle routine distinguishes two intersections, tangency, separation, containment, coincidence, concentricity, and invalid radii. Each candidate is reconstructed through the general two-point transform and checked against ordinary joint ROM inequalities before acceptance.

When two candidates remain, the selected branch minimizes distance to the preceding valid world position of joint X . This history is a transient map keyed by arbitrary joint ID. If no valid snapshot exists, the current initial joint position supplies the preference. Analytic steps retain their circle centres, radii, candidates, and selected point for optional construction rendering; rendering only reads this trace.

9 Singularity and constraint diagnostics

A tangent dyad is a branch-merging analytic singularity. Coincident circles are degenerate and underdetermined rather than a unique closure. Independently, a drop in the locally expected Jacobian rank marks a singular or near-singular component. Zero mobility by itself is not singularity.

Three other conditions remain distinct. For m scalar residuals and local rank r :

- a structural overconstraint flag identifies a subsystem with more required scalar equalities than available variables;
- $m - r > 0$ identifies dependent or redundant residual rows at the sampled pose, which need not invalidate a consistent mechanism; and
- geometric inconsistency means the closure norm cannot be reduced below tolerance.

Diagnostics report component membership, passive and driven mobility, rank, unresolved links, residual norm, analytic solve count, numerical-fallback use, and explanatory messages. Disconnected or intentionally free components may therefore be valid while still carrying useful warnings.

10 Bounded numerical fallback

After analytic propagation reaches a fixed point, rank analysis decides whether an unresolved subsystem is genuinely free or expected to close. Free coordinates preserve their finite prior values and do not enter an optimizer. For a constrained remainder, the initial condition combines the current partial analytic result with the preceding valid coordinates. A damped Gauss–Newton/Levenberg–Marquardt step solves

$$\left(J^T J + \lambda D \right) \Delta \mathbf{q} = -J^T \mathbf{r}, \quad (22)$$

where D provides diagonal scale regularization. Central differences construct J ; translation and angular increments have separate caps. A bounded line search accepts only finite, residual-decreasing candidates. Ordinary joint ROM bounds use constant-dimension active-limit residuals together with simultaneous averaged angle projection, avoiding constraint-ID ordering as a numerical policy. The actuator revolute instead follows the servo’s absolute command bounds.

The iteration has a finite cap, exits early below closure tolerance, and reports failure rather than committing a nonfinite or high-residual pose. Numerical minimization is thus a local kinematic fallback, not dynamics, branch search, or a device for hiding mobility.

11 Simultaneous finger and contactor solve

The finger remains a serial proximal–middle–distal chain downstream of the general mechanism graph. DIP flexion is coupled to PIP flexion by $\gamma = k\beta$, with $k = 0.62$. For contactor j , let $\mathbf{F}_j(\alpha, \beta)$ be its point on the finger centreline and $\mathbf{n}_{d,j}$ the segment normal directed toward positive world Y . Let W_j be total internal ring width. Its admissible domain and per-side affordance are

$$W_j \geq w_{f,j}, \quad a_j = \frac{W_j - w_{f,j}}{2} \geq 0. \quad (23)$$

The linkage centreline target is

$$\mathbf{T}_j = \mathbf{F}_j + \left(\frac{W_j}{2} + \frac{w_{l,j}}{2} \right) \mathbf{n}_{d,j}. \quad (24)$$

The specialized default solve uses $\mathbf{z} = (\alpha, \beta, \eta_m, \eta_d)$: MCP and PIP angles plus the two passive driver angles. With linkage contact points $\mathbf{P}_m$ and $\mathbf{P}_d$, it minimizes

$$\mathbf{r}_c(\mathbf{z}) = \begin{bmatrix} \mathbf{P}_m - \mathbf{T}_m \\ \mathbf{P}_d - \mathbf{T}_d \end{bmatrix}. \quad (25)$$

A projected damped iteration enforces hand and driver limits. If one variable reaches a bound, available variables continue solving. This stage consumes the already-resolved upstream linkage and does not invoke a four-bar solver.

Every rectangular link is then tested against each phalanx capsule, except for the link carrying the contactor assigned to that phalanx. For other pairs, centreline segments S_l, S_f , widths w_l, w_f , and clearance ϵ , the pose requires

$$\text{dist}(S_l, S_f) \geq \frac{w_l + w_f}{2} + \epsilon. \quad (26)$$

Ground-plane and dorsal-rail crossings are also rejected. Failure restores the last valid mechanism, hand, mount, contactor, and joint-limit state.

12 Gravity-only static moments

Each phalanx is approximated as a uniform cylinder, consistent with published index-finger and hand parameterizations [7, 9]. For rendered length L_i , diameter d_i , and tissue density $\rho = 1160 \text{ kg m}^{-3}$, mass is

$$m_i = \rho \pi \left(\frac{d_i}{2} \right)^2 L_i. \quad (27)$$

With gravity $\mathbf{g} = (0, -g)$ and joint point $\mathbf{J}_k$, the world gravity moment from distal segments is

$$\tau_k^{g,z} = \sum_{i \in \mathcal{D}(k)} (\mathbf{r}_{i,k} \times m_i \mathbf{g})_z, \quad \mathbf{r}_{i,k} = \mathbf{C}_i - \mathbf{J}_k. \quad (28)$$

An optional torsional term has form $\tau_k^s = -K_k(q_k - q_{0,k})$, with default $K_k = 0$, and the displayed holding moment is $\tau_k^{hold} = -(\tau_k^g + \tau_k^s)$. It excludes tendon forces, contact forces, friction, and joint reactions [8].

13 Interaction, construction geometry, and export

Dragging the driven-link handle converts the direction from its arbitrary servo attachment to the visual handle into the link’s absolute angle. Other actuator-connected handles sample the finite primary-servo range and retain the valid pose with the nearest endpoint. A genuinely free attached link rotates about its revolute point. This remains a bounded one-coordinate inverse control rather than a general drag constraint.

The joint inspector exposes both graph endpoints: the reference may be world or another link, and the target is a link. Reassigning either endpoint preserves the current world hinge while recomputing the corresponding local attachment coordinates, after which the ordinary graph build and validation path runs. Self-edges and missing targets are rejected.

The construction layer retains ordinary linkage cues and consumes the latest analytic trace. For each discovered dyad it can render both circle loci and the selected shared-joint intersection. These values originate in the simulation stage; Canvas rendering remains read-only.

The JSON exporter serializes a schema name and version, timestamp, units, shared hand scale, active tab, and detached D2–D5 persistent states. Solver diagnostics, analytic trace, and prior-joint branch history are transient and are omitted. Angles remain radians and lengths remain millimetres.

14 Update ordering and invariants

The frame update is ordered as follows:

1. advance or accept the absolute servo command;
2. synchronize editable dorsal mounting geometry;
3. build and analyze the constraint graph;
4. run analytic propagation and, only when required, numerical fallback;
5. solve the specialized finger/contact constraints;
6. validate collisions and hard geometric exclusions;
7. calculate finger statics;
8. capture the valid finite snapshot and render it.

Every committed revolute satisfies Equation (14) within named closure tolerance and every non-actuator revolute satisfies its configured ROM inequality. All link positions, angles, and residuals must be finite. Analytic branches use only prior valid joint history, and a failed candidate or topology edit cannot replace that snapshot.

15 Remaining mathematical scope

The graph presently models revolute planar mechanisms, bounded linear slots, and one prescribed rotational servo. Slot friction is metadata only: there is still no force-equilibrium or stick-slip solve. The model has no general prismatic actuator, multiple independent actuator commands, compliance, rigid-body dynamics, or force-balanced contact. The numerical fallback is local and does not guarantee a global solution or deliberate assembly-branch transition. The two finger contactors remain a specialized downstream solve rather than arbitrary graph constraints.

The next coherent milestone is to replace the demonstrator geometry with a calibrated reference-derived candidate, express its single distal contact and finger coordinates in the same residual formulation, and wrap the implemented local sensitivities in a bounded multi-pose optimizer. Force-balanced slot reactions and experimentally calibrated friction belong in a subsequent quasi-static stage.

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